

Genealogy Dynamics: A Graph-Based Two-Sex Cluster-Dynamics Model of the U.S. Population, 1650–2050

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Adapted from the graph-based cluster-dynamics framework (RadCluster).

Outline

Graph-Based Population Dynamics

Model Formulation

Reaction Kernels and Input Data

Results

Future Work and References

Related work

- ▶ **Pedigree theory:** Chang (1999) — two-parent Wright–Fisher, genealogical MRCA at $\sim \log_2 n$, identical-ancestors point at $1.77 \log_2 n$; Rohde, Olson & Chang (2004, *Nature*) — recent common ancestry survives realistic substructure (simulation); Derrida, Manrubia & Zanette (1999, 2000) — pedigree-collapse statistics, stationary after $\sim \log N$ generations.
- ▶ **Cultural transmission:** Manrubia & Zanette (2002) — birth–death model of a vertically transmitted trait (surnames); Cavalli-Sforza & Feldman (1981) — quantitative theory of cultural transmission, foundation for group endogamy + disaffiliation.
- ▶ **Caveat:** Matsen & Evans (2008) — genealogical $\neq$ genetic ancestry; cluster depth is about pedigrees, not genes.
- ▶ **This work:** forward-time cluster-dynamics ODEs ported from irradiated materials; catalytic mating on an explicit RAG; two-sex, age, regional, religious structure; 1650–2025 U.S. run on measured series + 2025–2050 prediction.

From clusters to genealogy clusters

- ▶ RadCluster partitions **defect clusters** into discrete classes (vertices); every reaction is an admissible transition (edge) with a rate kernel.
- ▶ The same mathematics describes a **human population**:
 - ▶ vertices $\rightarrow$ clusters $C_0, \dots, C_n$ in two populations $P_{\text{pat}}, P_{\text{mat}}$ ($n=14$ default),
 - ▶ edges $\rightarrow$ immigration, mating/birth, death, emigration.
- ▶ State $\mathbf{c}(t) \in \mathbb{R}_{\geq 0}^{2(n+1)}$ (millions): paternal block then maternal block, integrated as ODEs.
- ▶ Two cases: **Case 1**, the 1650–2025 colonial run on measured inputs; **Case 2**, a 2025–2050 projection from the Case-1 handoff state.
- ▶ Two inheritance rules carried in parallel:
 - ▶ **Shallowest inheritance**: $\text{child} = \min(n, 1 + \min(q, r))$ — the child takes the *shallower* parental bloodline; deep lineages are pulled down in mixed pairings.
 - ▶ **Deepest inheritance**: $\text{child} = \min(n, 1 + \max(q, r))$ — the child takes the *deeper* parental bloodline; depth never decreases, so mass ratchets to the cap.
- ▶ Assortativity α : **0.8** enforced central (anchored to mean religious endogamy $\bar{\rho}_g \approx 0.78$); 0.0 null, 0.9 bound. CPS-calibrated 0.66.

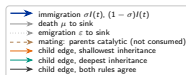
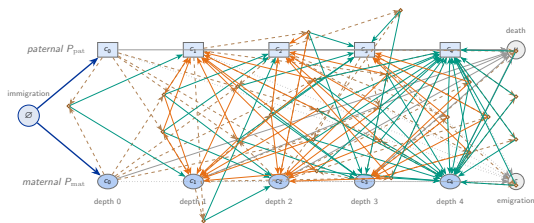
Master equation (matrix form)

State $\mathbf{c}(t) \in \mathbb{R}^{2(n+1)}$: $c_{d,\text{pat}}$ = paternal cluster C_d , $c_{d,\text{mat}}$ = maternal (millions; $n = 14$ default).

$$\frac{d\mathbf{c}}{dt} = \mathbf{P}(t) + \mathbf{S} \mathbf{J}(\mathbf{c}) - \mathbf{D}(t) \mathbf{c}$$

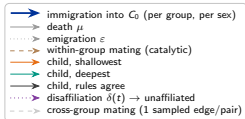
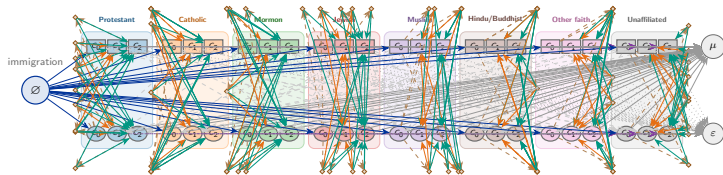
- ▶ **Source:** $\mathbf{P}(t)$ — immigration $I(t)$ splits $\sigma/(1-\sigma)$ into C_0 of each population ($\sigma(t)$ data-driven series, Step 5).
- ▶ **Loss:** $\mathbf{D}(t)$ diagonal — mortality $\mu(t)$ plus emigration ε , uniform across sexes (default); sex-specific rates supported.
- ▶ **Flux:** $J_{qr} = B [\alpha \delta_{qr} m_q^{\text{pat}} + (1 - \alpha) m_q^{\text{pat}} m_r^{\text{mat}}]$ over ordered pairs (q, r) ; $B = \min(\sum \beta^{\text{pat}} c_{\text{pat}}, \sum \beta^{\text{mat}} c_{\text{mat}})$ (marriage-function style).
- ▶ **Stoichiometry:** $S_{(d,\text{pat}),(q,r)} = s$, $S_{(d,\text{mat}),(q,r)} = 1-s$ iff $\text{child}(q, r) = d$ ($s = 0.512$ measured, NCHS [7]) — each $\mathbf{S}$ column sums to 1, so births conserve people exactly.
- ▶ Parents are catalytic (not consumed). Extensions: $\times 8$ religious groups (240 ODEs at $n=14$), $\times 18$ age bands (540 states), $\times 4$ regions (120 states).

Reaction graph: two-sex model



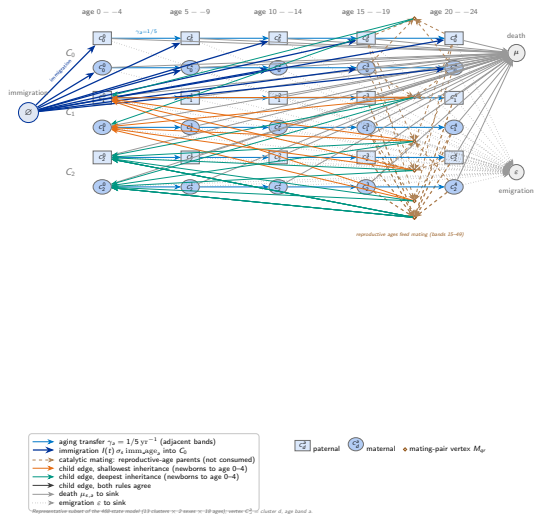
Generated *from the model code* (NetworkX, seed 20260926): every vertex a state variable, every edge a right-hand-side term. Rectangles paternal, ellipses maternal; diamonds M_{qr} mating pairs; child edges colored by rule agreement.

Reaction graph: religious extension



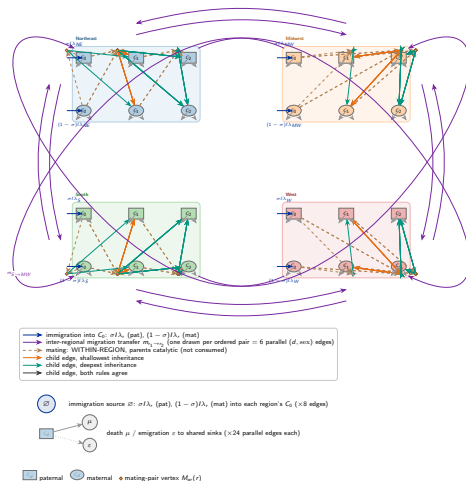
$n=2$ shown (240 ODEs at $n=14$): group-colored lanes, within-group mating, sampled cross-group edges, disaffiliation transfers $(d, s, g) \rightarrow (d, s, \text{una})$ at $\delta(t)$.

Reaction graph: age-structured model



$n=2$, ages 0-4 to 20-24 shown (540 states): aging edges at $\gamma_a=1/5/\text{yr}$, reproductive-age mating, newborns into band 0.

Reaction graph: regional model



$n=2$ shown (120 states): within-region mating diamonds; one transfer arrow $m_{r_1 \rightarrow r_2}$ per ordered region pair; immigration splits by region into C_0 .

Reaction kernel table

#	Process class	Transition	Kernel / flux
1	Immigration (source)	$\emptyset \rightarrow (C_0, \text{pat}), (C_0, \text{mat})$	$\sigma(t)I(t), (1-\sigma(t))I(t)$
2	Birth (mating)	$((C_q, \text{pat}), (C_r, \text{mat})) \rightarrow (C_g, \cdot)$	$J_{qr} = B [\alpha \delta_{qr} m_q^{\text{pat}} + (1 - \alpha) \delta_{qr} m_r^{\text{mat}}]$ $B = \min(\sum \beta^{\text{pat}} c_{\text{pat}}, \sum \beta^{\text{mat}} c_{\text{mat}})$ births split $s/(1-s)$ into
3	Mortality (sink)	$(C_d, \alpha) \rightarrow \emptyset$	$\mu_\alpha(t) c_{d,\alpha}$
4	Emigration (sink)	$(C_d, \alpha) \rightarrow \emptyset$	$\varepsilon_\alpha c_{d,\alpha}$

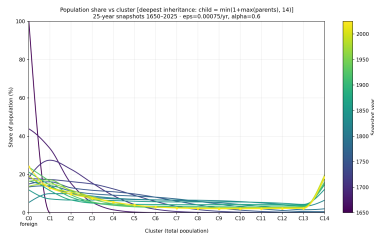
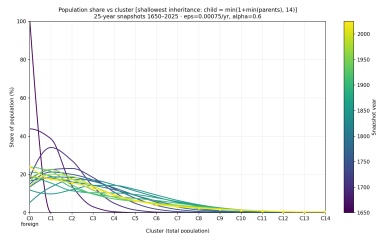
Child-cluster kernel $g(q, r)$: shallowest $\min(n, 1 + \min(q, r))$; deepest $\min(n, 1 + \max(q, r))$. Parents are catalytic.

Input parameters and assumptions

Parameter	Symbol	Value / series
LPR immigration	$I_{LPR}(t)$	FY1980–FY2024
Unauthorized inflow	$I_{unauth}(t)$	1980–2024
Colonial immigration	$I(t)$	1650–1819
Fertility	$\beta(t)$	$2 \text{ CBR}(t) / 1000$ (uniform sexes)
Mortality	$\mu_{\alpha}(t)$	$\text{CDR}(t) / 1000$ scaled by the measured HLD sex differential (2024 ≈ 1.069)
Emigration	ε	$7.5 \times 10^{-4} / \text{yr}$ (uniform)
Sex ratio at birth	s	0.512 (≈ 1.05 male/female)
Immigrant M fraction	$\sigma(t)$	0.65 ass. (1650–1819); 0.629 est. (1820–1917); $\rightarrow 0.50$ by 1950 est.; DHS Tbl. 9 meas. + unauth. p.
Assortativity	α	0.8 central; swept 0.0, 0.8, 0.9
Age fertility/mortality	$\beta_a(t), \mu_{s,a}(t)$	18 bands; 1900–2023
Inter-regional migration	$m_{r \rightarrow r'}$	12 pairs, 2023–24
Immigration allocation	$\lambda_r(t)$	4 regions, FY2014–23
Case-2 β, μ, I, δ	projected	CBR 10.26 $\rightarrow$ 5.26; CDR 9.20 $\rightarrow$ 11.09/1000; $I=1.705\text{M}/\text{yr}$

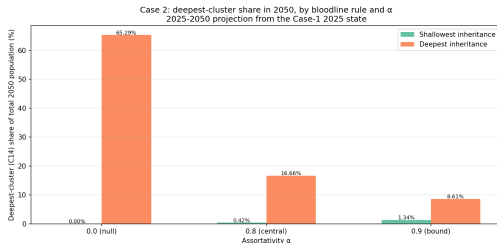
Validation: 1980–2024 hindcast within **2.87 M max error** (-0.8% in 2024); exact total-balance to 10^{-15} . Every input labeled measured / interpolated / estimated / calibrated / assumed / projected.

Case 1: colonial run 1650–2025 — the rules diverge



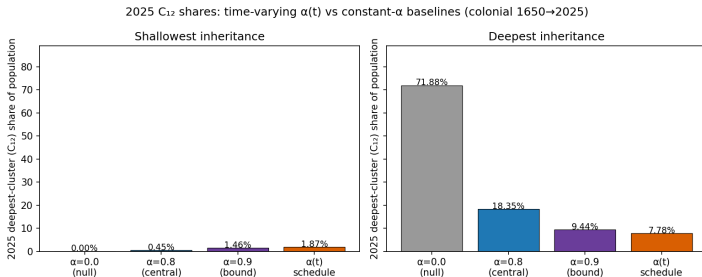
- (a) Shallowest: $C_{14} = 0.45\%$ by 2025 ($\alpha=0.8$) (b) Deepest: $C_{14} = 18.35\%$ by 2025 ($\alpha=0.8$)
2025 C_{14} share, $\alpha=0.0/0.8/0.9$, $n=14$: shallowest 0.00/**0.45**/1.46%; deepest
71.88/**18.35**/9.44%. Totals rule-independent (259.0 M). **Mechanism:**
shallowest pulls deep lineages down in mixed pairings; deepest never pulls
down, so depth ratchets to the cap. Stronger α *reduces* the deepest
accumulation (shallow groups pair shallow).

Case 2: 2025–2050 projection



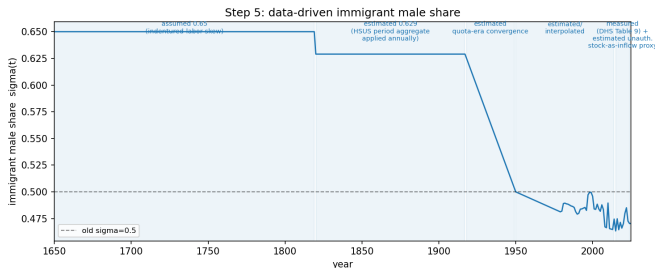
From the Case-1 2025 handoff state. 2050 total: **279.7 M** (both rules). 2050 C_{14} at $\alpha=0.8$ (2025 in parens): shallowest **0.42%** (0.45%); deepest **16.66%** (18.35%). Full 2050 α sweep (0.0/0.8/0.9): shallowest 0.00/0.42/1.34%; deepest 65.29/16.66/8.61%. Both rules dilute slightly as post-2025 immigration refreshes C_0 (24.0% $\rightarrow$ 30.2%). Inputs: $\beta(t)$ OLS-projected CBR, $\mu(t)$ OLS-projected CDR (excl. COVID), I frozen at 2024, σ held at 2024 blend (0.4703), ε calibrated.

Time-varying assortativity $\alpha(t)$



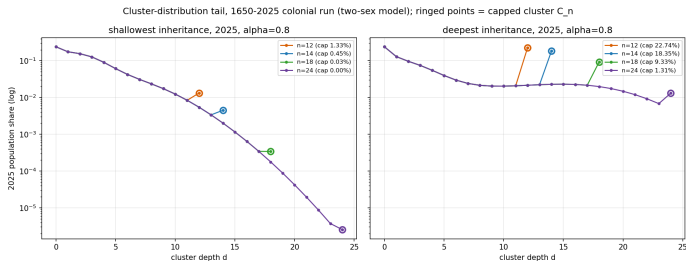
The $\alpha(t)$ schedule (0.95 colonial $\rightarrow$ 0.653 in 2025, from intermarriage literature and Pew series [?]) gives 2025 C_{14} of **1.87%** shallowest / **7.78%** deepest, vs 0.45% / 18.35% at constant $\alpha=0.8$. Shallowest retains path dependence (exceeds the 0.9 bound); deepest undershoots (schedule ends below 0.8 when recent mixing matters most). Sensitivity only — constant 0.8 stays central.

Sex-specific rates: the maternal bottleneck



Immigrant male share $\sigma(t)$: 0.65 *assumed* (1650–1819); 0.629 *estimated* (1820–1917, HSUS [9]); $\rightarrow 0.50$ by 1950; measured DHS LPR-by-sex [10] + unauthorized proxy [11] from 2015. Male-skewed immigration + $\min(W^{\text{pat}}, W^{\text{mat}})$ makes the maternal side binding: 2025 total 259.0 M (old σ 300.2 M); anchor worst error -29.3% (1830) — reported, not recalibrated. Mortality sex differential measured (HLD [14], 2024 ≈ 1.069).

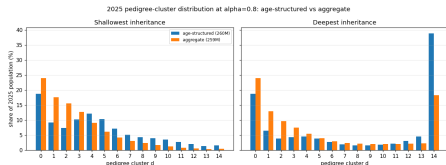
Depth resolution: $n=14$ and the bit-conserved cap



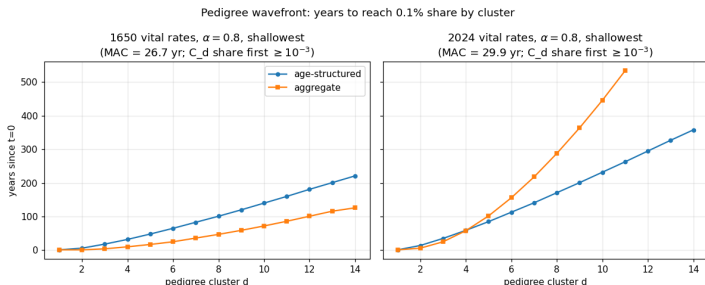
2025 C_n at $\alpha=0.8$: shallowest 1.33/**0.45**/0.03/0.00%; deepest 22.74/**18.35**/9.33/1.31% ($n=12/14/18/24$). Totals n -invariant (258.95 M); cap bit-conserved ($C_{14} = \sum_{d \geq 14}$) under both rules. Deepest cap accumulates by design (ratchet, not pile-up); $n=24$ resolves the tail (flat $\sim 2.2\%$ /cluster to C_{17}). $n=14$ is the operating point.

Age structure: resolved tail nearly matches the aggregate

- ▶ 540 states: $15 \times 2 \times 18$ ($n=14$); aging $\gamma=1/5/\text{yr}$; $\sigma(t)$ series (Step 5).
- ▶ 2025 C_{14} at $\alpha=0.8$: shallowest **1.62%** vs 0.45% agg.; deepest **38.90%** vs 18.35% agg. — age structure *amplifies* deep concentration, especially under deepest inheritance: older cohorts preserve a deep reservoir that the ratchet then lifts to the cap.
- ▶ Male-fertility sensitivity (age model): fathers ~ 2 yr older $\rightarrow$ 2025 C_{14} -0.8 pp (shallowest) to -10 pp (deepest) at $\alpha=0.8$; totals -4.9% — material, stays *assumed*.

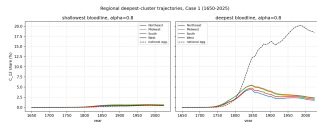
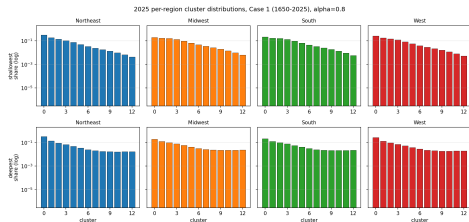


Generation-delay wavefront



Years for each cluster to reach 0.1% share (closed population, constant rates, shallowest inheritance, $\alpha = 0.8$). Left (1650 rates): aggregate climbs $\approx 1.9\times$ faster — missing generation-delay artifact (no 15-yr parental-age floor). Right (2024 rates): aggregate is mass-accumulation-limited and runs slower. A *rate* measurement, independent of n .

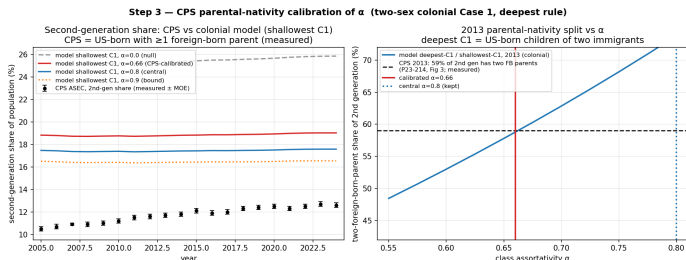
Regional compartments: the Midwest runs deepest



C_{14} trajectories 1650–2025

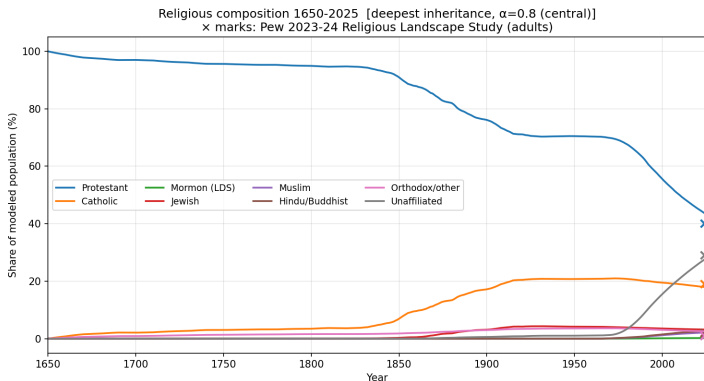
2025 per-region cluster distributions (Case 1, $\alpha=0.8$)
120 states, 4 Census regions; within-region mating (assumed); migration conservative. 2025 C_{14} ($\alpha=0.8$): shallowest NE 0.36 / MW **0.53** / S 0.49 / W 0.43%; deepest NE 14.74 / MW **20.61** / S 19.54 / W 17.20%. Midwest deepest under both rules: smallest immigration share (11.3%, DHS Tbl. 4 [2]) dilutes deep mass least; flows from ACS 2023–24 [1]. Regional Case-2 projection in progress.

CPS parental-nativity calibration of α



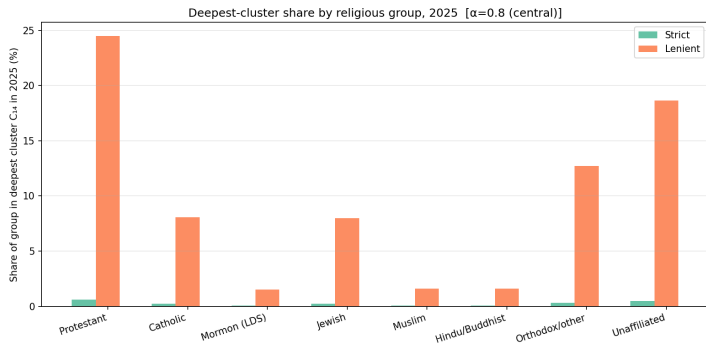
CPS ASEC 2024 Tbl. 1: second-generation share 2005–2024, 10.5%→12.6% (measured [3]); P23-214: 2013 second generation 11.7%, 59% with two foreign-born parents (measured [4]). Model ratio deepest- C_1 /shallowest- C_1 vs 59% → $\alpha_{\text{CPS}}=0.66$; $|0.66-0.8|=0.14$ not material → keep central $\alpha=0.8$ (also anchored to mean religious endogamy 0.78).

Religious dimension: 8 groups, $n=14$



State (d, s, g) : 240 ODEs. Group endogamy ρ_g : mor .87, pro .81, cat .75, mus .84, jew .62, una .68 (est. from Pew), dhr .85, oth .80 (ass.); children take the **mother's** group (ass.). Disaffiliation ramps 1970–90 to $\delta_{\max}=8\times 10^{-3}/\text{yr}$, **calibrated** to Pew unaffiliated 29% (model 27.9%, rule-independent). $\times$ = Pew 2023–24 RLS anchors [9]. 2025 composition ($\alpha=0.8$, both rules): pro 43.3 / cat 17.9 / mor 0.2 / jew 3.2 / mus 2.2 / dhr 2.7 / oth 2.6 / una 27.9%.

Deepest cluster C_{14} by religious group, 2025



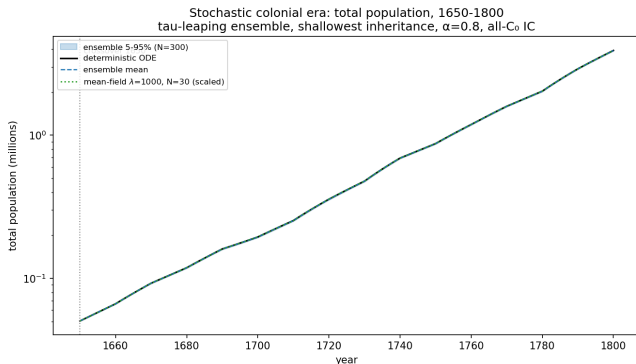
C_{14} share of total 2025 pop. ($\alpha=0.8$, $n=14$): shallowest 0.45% / deepest **17.93%**. Demographic, not theological: long-tenure groups run deep (Protestant 24.5%, unaffiliated 18.7% under deepest); post-1965-immigrant groups stay shallow (Muslim, Hindu/Buddhist $\approx 1.6\%$). Under shallowest, all groups sit below 0.7%.

Case-2 religious projection, 2025–2050

Run in progress under the new inheritance rules.

The 2025–2050 religious projection continues disaffiliation at its calibrated rate from the Case-1 2025 handoff state. Prior pattern to retest: the unaffiliated share climbs on the one-way disaffiliation flow (a flagged artifact; two-way switching is future work).

Stochastic colonial era: the deterministic trajectory is unbiased



- ▶ τ -leaping, $N=300$ traj. 1650–1800, both rules.
- ▶ No bias: $|z| \leq 1.40$, max dev 6.1×10^{-4} ; mean-field check 7.7×10^{-5} .
- ▶ Deep mass robust: $CV(C_{14})$ 9.5% (shallowest) / 2.7% (deepest) at 1800.
- ▶ $CV(\text{total})$ peaks 0.72% ~ 1710 , never reaches 1%; 0/300 empty C_{12} – C_{14} .

Independent validation: rerun in progress

Validation checks being rerun under the new rules.

- ▶ **(a) “American”-ancestry** (2023 ACS B04006, measured [16]): regional deep-share ranking vs observed — prior run found a structural gap (model MW-peaked vs observed S-peaked, $\rho=0.60$); retesting with shallowest/deepest.
- ▶ **(b) Surnames:** genuine-attempt negative — no defensible check found (2010 list [17]); stands.
- ▶ **(c) Foreign-born** (B05002; consistency, not validation): level gap decomposition into the total bias + unauthorized accounting; rank correlation to retest.

Suggestions for more accurate models and data

1. **Age $\times$ religion interaction:** the 540-state age model and 240-ODE religious model are verified unbroken but not yet combined.
2. **Regional vital-rate differences:** regional fertility/mortality and cluster-specific migration unmodeled; regional totals uncalibrated.
3. **Paternal age schedule:** male fertility still the female ASFR shape (assumed); a ~ 2 yr older paternal schedule moves 2025 C_n by 2–5 pp at $\alpha \geq 0.8$; calibrate from NCHS fatherhood data.
4. **$\alpha(t)$ for the extended models:** the time-varying assortativity schedule is two-sex only; port it to the religious, age, and regional models.
5. **Re-affiliation in disaffiliation:** the one-way flow is an honest artifact (unaffiliated 27.9% $\rightarrow$ 36.4% by 2050); add re-affiliation.
6. **Settlement-history process for regional deep shares:** Step 8 found a real structural gap (model peaks MW, observed “American” ancestry peaks S) — add a settlement-history/identity mechanism before the regional deep ranking can be trusted.

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Thank you